Jeffrey Tom Shaji

Recent Computer Science Engineering graduate and eager Machine Learning enthusiast with hands-on experience building RAG pipelines using LangChain, Hugging Face, and Ollama. Actively pursuing the further specialization in AI to deepen expertise in neural networks and contribute to the rapidly evolving field of AI.

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WORK EXPERIENCE

OWL CYBER DEFENSE

Engineering Intern — Aug 15 - Sept 16 2023

- Configured and tested OWL data diodes for secure data transfer
- Conducted packet transfer simulations using both TCP and UDP protocols.
- Configured and connected servers that run on EsXi to OWL data diodes and successfully executed data transfers.

EDUCATION

Vellore Institute of Technology Chennai

Bachelor of Technology in Computer Science and Engineering - 8.76 — Aug 2021 - May 2025

PROJECTS

Retrieval-Augmented Generation (RAG) using LangChain

- Implemented a RAG pipeline using Ollama for text embeddings and Groq for the large language model to enable semantic document retrieval and question answering.
- Utilized PDF document loaders and vector stores to build a custom knowledge base from a corpus of technical documentation and textual data.
- Designed and tested prompt templates and retrieval chains to improve response accuracy and relevance.

Next Word Prediction Model

- Developed a Next Word Prediction LLM from scratch, implementing custom preprocessing, Byte Pair Encoding tokenization, and text generation pipeline.
- Integrated GPT-based pretrained weights to enhance prediction accuracy and language fluency.
- Built and optimized the model for real-time text suggestion, showcasing understanding of transformer architectures and transfer learning techniques.

SKILLS

- Mobile Development: Flutter (Cross-Platform: Android & iOS)
- Backend & Databases: SQL, Firebase
- AI/ML: LLMs, Neural Networks, Deep Learning, LangChain
- Blockchain: Zero-Knowledge Proofs (ZKPs), PLONK
- Tools: Git, Canva

CERTIFICATE

Internship at OWL CYBER DEFENSE,U.A.E

The Complete Flutter Development BootCamp with Dart

Introduction to Generative AI

Introduction to Large Language Models

Neural Networks and Deep Learning – Coursera (DeepLearning.AI)

PUBLICATION

RangeShield ZKP - IGI Global

- Collaborated with a team to design and implement RangeShield ZKP, a novel Zero Knowledge Proof system that ensures privacy and correctness in federated learning on sensitive healthcare data.
- Integrated Pedersen Commitments to create binding, non-reversible proofs of model updates, preventing data corruption.

Link: https://www.igi-global.com/article/rs-zkp/394815